

Math 61: Introduction to Discrete Structures
Discussion 5: Counting Methods: Basic Principles and Permutations
and Combinations.

1 Counting Principles

1. How many positive divisors does 2000 have? (Hint: $2000 = 2^4 \cdot 5^3$)
2. The following questions deal with the number of relations on an n -element set
 - (a) How many symmetric relations are there on an n -element set? ([Solution](#))
 - (b) How many antisymmetric relations are there on an n -element set. ([Solution](#))
3. Two distinct dice are rolled. How many outcomes give the sum of 2 or the sum of 12?
4. An eight-bit string is a string of 0's and 1's that is of length 8, (e.g. 11010100) is an eight-string). Determine how many strings start with a 1 or end with a 1 or both. (Hint: Inclusion-Exclusion Principle)
5. Prove the Inclusion Exclusion Principle for three finite sets

$$|X \cup Y \cup Z| = |X| + |Y| + |Z| - |X \cap Y| - |X \cap Z| - |Y \cap Z| + |X \cap Y \cap Z|$$

(Hint: Use the two set Inclusion-Exclusion Principle with X and $Y \cup Z$ as your two sets)

2 Combinations and Permutations

1. How many 5 letter passwords can be generated using the following symbols without repetition, $a, b, \dots, z$, $A, B, \dots, Z$, and $0, 1, 2, \dots, 9$.
2. We toss a coin ten times
 - (a) How many outcomes have exactly three heads?
 - (b) How many outcomes have at most three heads?
 - (c) How many outcomes have as many heads as tails
3. Show that the product of any positive integer and its $k - 1$ successors is divisible by $k!$
4. If X is an n -element set and Y is a m -element set, $n \leq m$, how many one-to-one functions are there from X to Y . If $n = m$, how many bijections?
5. Prove that the number of solutions to the equation

$$x_1 + x_2 + x_3 = n \quad (n \geq 3)$$

where $x_1, x_2, x_3 > 0$ is $\binom{n-1}{2}$. (Hint: [Can use Stars and Bars](#))

6. Prove that the number of solutions to the inequality

$$x_1 + x_2 + \dots + x_n \leq M$$

where all variables are non-negative integers is $\binom{M+n}{n}$ ([Solution](#))